

Introduction to Concurrency

- **Concurrency:** Multiple execution flows make progress at the same time by interleaving their executions or by executing instructions at exactly the same time
- **Parallelism:** Multiple tasks running simultaneously, at the exact same time
- Hardware threads dictate the amount of parallelism through parallel execution of multiple kernel threads
- Steps for Parallelisation
 - Decomposition of computation
 - Assigning tasks to threads
 - Orchestration: Communication, synchronisation, scheduling
 - Mapping of threads to physical cores
- **Race Condition:** Timing or order of events affects outcome of executing concurrent code, not necessarily a flaw
- **Data Race**
 - Multiple threads access a shared resource without any synchronisation AND at least one thread modifies it, causing UB
 - Does not occur if both evaluations are on the same thread, both are atomic, or one of them happens before another
- Use mutual exclusion to synchronise access to shared resources
- Other Mechanisms
 - Locks: Primitive with minimal semantics, used as a building block
 - Semaphores: Basic, easy to get the hang of but hard to program with
 - Monitors: High level, requiring language support, implicit operations
 - Messages: Simple model of communication based on atomic transfer of data across a channel
- **Deadlock:** Execution where no progress is made when it should be possible
- Can either ignore it, prevent it, avoid it, or detect and recover from it
- **Starvation:** Thread is prevented from making progress because other threads have resource it requires and initial thread cannot acquire it, usually side effect of scheduling algorithm

Threads and Synchronisation in Modern C++

- Code to Execution:
 - Compilation and linking: Done by compiler
 - Loading: Loader is usually specific to OS
 - Execution: Coordinated by OS, program gets access to resources
- C++ Building:
 - Preprocessor replaces directives
 - Compiler parses C++ code and converts it to assembly
 - Assembler assembles code into machine code
 - Linker produces final compilation output from object files
- **Lifetimes**
 - Begin when storage with proper alignment and size for its type is obtained, and initialisation is complete
 - Ends when object is destroyed, or when destructor call starts, or the storage which the object occupies is released
 - Lifetime of a reference begins when initialisation is complete and ends as if it were a scalar object
- **Ownership**
 - Owner is an object containing a pointer to an object allocated by new for which a delete is required
 - Objects on the free store must have exactly one owner
 - Thread instances own a resource, and instances of threads are movable but not copyable

- **Structure of Program Data**
 - Every variable is an object
 - Every object occupies at least one memory location
 - Adjacent bit fields are different memory locations but can be part of the same storage unit
- **Invariants:** Statements always true about a particular data structure (might be broken during update)
- Wrap data structure with protection mechanism to prevent data races
- Mutexes provide mutual exclusion and serialisation where threads take turns to access the protected data
- Recommended usage: `std::lock_guard` with `RAII` to unlock on destruction
- Don't pass pointers / references to protected data outside the scope of the lock
- Avoid nested locks and calling user-supplied code while holding a lock; acquire locks in a fixed order
- **Condition Variable**
 - Associated with event or other condition; one or more threads can wait for condition to be satisfied
 - When condition is satisfied, thread notifies one or more threads waiting on condition variable to wake up
 - Condition check should not have side effects
 - **Spurious Wake:** Waiting thread reacquires mutex and checks condition even though it is not notified

Atomics and Memory Model in C++

- Correctly synchronised program should behave as if memory operations are executed in an order that is equivalent to some sequentially consistent interleaved execution of memory operations of each thread, where each write appears to be atomic and globally visible to all processors
- Processor might reorder operations to improve efficiency
- Memory models provide a contract to programmers about what are the visible side effects of their memory operations
- **As-if Rule**
 - The compiler is permitted to perform changes to the program as long as
 - Accesses to volatile objects occur strictly according to the semantics of the expressions in which they occur
 - They are no reordered wrt other volatile access on the same thread
 - At program termination, data written to files is exactly as if the program was simply executed
 - Prompting text which is sent to interactive devices will be shown before the program waits for input
 - Programs with UB are free from this rule

Atomics

- `std::atomic` allows for low-level synchronisation, with operations that reduce to 1 or 2 CPU instructions
- Atomic operations are indivisible and always observed fully done/undone
- Atomic loads will retrieve either initial value of object or value stored by some modification
- A non-atomic operation might be seen as half-done by another thread
- Atomic types provide `is_lock_free()` which checks if operations on a given type are done with atomic instructions, which might be emulated with internal mutexes

Modification Order

- Consists of all writes to an object from all threads in the program, can vary
- With atomic variables, compiler ensures threads agree on MO for each variable, but might still have races
- Once a thread has seen an entry, subsequent reads and writes must return that entry or a later one in the MO
- Threads do not have to agree on separate objects

Memory Order

- C++ Memory Order: Specifies how memory accesses are to be ordered around an atomic operation

Sequentially Consistent Ordering

- Behaviour of program is consistent with a simple sequential view of the world
- All threads must agree on the same order of operations
- Operations within same thread can't be reordered
- Sequentially consistent store synchronises with sequentially consistent load of same variable that reads the value stored
- Performance penalty on weakly-ordered machine with many cores due to need for synchronisation

Building Memory Model

- **Sequenced Before (sb):**
 - Within same thread, A is sequenced before B and will evaluate first
- **Synchronises With (sw):**
 - Appears between atomic load and store operations
 - Suitably tagged write / store / release in thread A on a variable synchronises with suitably tagged read / load / acquire in another thread if acquire operations reads value of that write, depends on exact execution
 - Can synchronise with `seq_cst` as well
- **Simply Happens Before (hb):**
 - A sb B
 - A sw B
 - A hb X, X hb B
- **Strongly Happens Before:**
 - Specifies which operations see the effects of other operations
 - If A shb B, then A appears to be evaluated before B in all contexts
 - A sb B
 - A sw B, A, B are seq cst
 - A sb X, X hb Y, Y sb B
 - A shb X, X shb B
- **Visible Side Effects:**
 - A side effect of A on a write M is visible to a read B
 - A hb B
 - There is no other side effect X such that A hb X, X hb B
 - The visible sequence of side effects is the longest contiguous subset of side effects to a scalar where B does not hb it
- All modifications to an atomic variable occur in a total order which guarantees coherence
 - WW: If evaluation A modifying atomic M hb B that modifies M, then A appears earlier than B in the MO for M
 - RR: If computation A of atomic M hb a computation B on M, and if the value of A comes from write X on M, then the value of B is either X or the value stored by a side effect Y on M that appears after X in the MO for M
 - RW: If computation A of atomic M hb operation B on M, then the value of A from a side effect X that appears earlier than B in the MO for M
 - WR: If a side effect X on atomic M hb a computation B of M, then the evaluation B takes its value from X or from side effect Y that follows X in the MO for M
- With non-seq cst orderings, there is no single global order of events, and threads only have to agree on MO of each variable

Relaxed Ordering

- Operations with relaxed ordering do not participate in sw relationships
- Operations on same variable within a single thread still obey hb
- Accesses to a atomic variable from the same thread cannot be reordered

Release-Acquire Ordering

- Use acquire and release memory ordering
- Atomic read-modify-write operations use either or both
- No total MO but introduces some synchronisation
- If a store in A with release and a load in B from the same variable with acquire and load in B reads a value written by a store in A, then the store in A sw the load in B
- All memory writes that hb the store in A are visible side effects in B; B must return value that A stored or a later value in the release sequence
- Other threads can still see different memory orders
- Release Sequence: The longest continuous subsequence of MO of M that consists of atomic RMW operations made to M after a release operation performed on it
- Sequential Consistency for data race free programs is guaranteed if every atomic uses seq cst

Concurrent Data Structures

- Goal: Multiple threads can access DS concurrently, and each thread sees a self-consistent view
- Thread-Safe DS should ensure no data is lost or corrupted, all invariants are upheld, and no problematic race conditions
- Mutexes are an easy way to provide serialisation but prevent true concurrent access
- Invariants are statements that are always true about a data structure, which might be broken during an update
- Constructors and destructors require exclusive access to DS; user has to ensure this and check whether DS can support assignment, swap, or copy construction
- Minimising the protected region results in fewer operations being serialised and greater potential for concurrency
- Locked-Based Concurrent DS: Ensure right mutex is locked when accessing and lock is held for minimum duration
- Shared Pointer: Holds a reference to pointed to data, keeps track of number of references (thread safe), automatically calling destructor once no longer referenced
- Fine-Grained Locking: Analyse constituent parts, associating one mutex with each distinct item, cannot continue using STL library
- To avoid deadlock, avoid nested locks, calling user supplied code while holding a lock, and always acquire locks in a fixed order
- Hand-Over-Hand Locking: Hold lock on current node, acquire next, release current, and continue; can cause deadlock on doubly-linked list

Lock-Free Concurrent Data Structures

- Blocking Data Structures: Use mutexes, condition variables, futures for synchronisation
- Calls library functions which suspend execution until another thread performs an action
- Non-Blocking Data Structures: Do not use blocking library functions
- Obstruction Free: If all other threads are paused, any given thread will complete its operation in a bounded number of steps
- Lock Free: If multiple threads are operating on a data structure, then after a bounded number of steps, one of them will complete its operation (no deadlock)
- Wait Free: Every thread operating on a data structure will complete its operation in a bounded number of steps even if other threads are also operating (no starvation)
- Lock freedom allows individual threads to starve but guarantees system-wide throughput
- An algorithm is lock free if when the program threads are run for a sufficiently long time, at least one of them makes progress

- All wait-free algorithms are lock-free
- Notion of Progress: Function returns, or effects become visible
- More than one thread must be able to access the data structure concurrently
- Wait-Free data structures: Does not result in thread being subject to starvation due to loops, does not involve unbounded number of retries
- Might end up writing spin lock in essence
- Benefits: Maximum concurrency, more robust
- Cons: Might cause livelock, and decrease overall performance due to atomic operations
- Contention and Cache Ping Pong: Changes to data need time to propagate to cache on other core, which might cause a stop and wait depending on memory ordering
- False sharing can produce cache ping pong as well when accessing different memory locations on the same cache line within multiple threads
- Guidelines
 - Prototype using sequentially consistent memory order
 - Use lock-free memory reclamation scheme
 - Watch out for ABA problem, prevalent with free lists (memory pools where nodes are recycled), solve using ABA counter
 - Identify busy-wait loop and help the other thread
- **ABA Problem**
 - Compare and swap checks equality before swapping
 - Suppose our data structure is modified before the CAS check, in a way such that the front happens to have the same address
 - The check passes, although the next pointer on the swapped in node points to garbage
 - To solve this, we can tag on generation counters so that even if the address is the same, the value is different
- **Use After Free**
 - If a thread gets preempted and the head gets deleted while it is trying to access, we might get a UAF
 - Can cause memory leak by not freeing anything
 - Mark nodes for deletion while threads are trying to pop and free all at once
 - Use reference counters and shared pointers
 - Use hazard pointers to track threads and their references
 - We choose to use a recycling center for nodes that should be "deleted", using a lock-free stack
 - This recycling center also needs synchronisation with the queue
- **Serialisation**
 - Due to preempting, it is possible to create a scenario where the actual operations performed do not correspond to a serialised interleaving
 - A pushes but gets interrupted, while B pushes in. B tries to pop, but since A is not finished, the queue still looks empty. Only after this try pop does A finish its work.

Testing and Debugging Multithreaded Programs

- Unwanted Blocking: Deadlock or livelock, or I/O and other external input
- Issues include data races, broken invariants, and lifetime issues
- Techniques for testing include stress testing, or special implementations for synchronisation primitives
- Test scalability and performance as well
- Useful tools for debugging memory issues include valgrind and sanitisers
- **Valgrind**
 - Memcheck uses shadow memory which tracks information on memory usage and is used to detect and report incorrect accesses
 - A-bit (Valid-Address): Corresponding byte accessible
 - V-bit (Valid-Value): Corresponding bit initialised
 - Ensure each allocation is freed only once, each access is to a currently allocated space, and all reads are to locations already written
 - Helgrind detects misuses of the pthread API, potential deadlock from lock

- ordering, and data races by intercepting function calls
- Builds directed graph of lock acquisition
- Checks order in which memory accesses happen, building DAG of happens before relationships which is used to flag cycles
- **Sanitisers**
 - Different types of sanitisers such as ASan (address), LSan (memory leak), TSan (thread), MSan (uninitialised memory), HWASAN, UBSan (undefined behaviour)
 - ASan uses shadow values to represent individual bytes allocation
 - Any aligned 8 bytes can have N good bytes and $8 - N$ bad bytes giving 9 states
 - TSan adds checks at every function entry and exit
 - It performs direct shadow mapping from actual addresses to shadow addresses
 - Each 8 byte shadow cell represents one memory access, storing TID, Epoch, Position in word, IsWrite
 - Uses this information to determines hb and races

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